ABSTRACT

CampusPulse: Smart Classroom Monitoring and Energy Management

There is often poor electricity management in schools. Classrooms may be empty, but fans and lights remain on. CampusPulse is a smart classroom and energy management system that uses the Internet of Things. It detects if people are in the classroom and adjusts electrical devices to save energy. This system relies on an ESP32 microcontroller, a PIR sensor to detect movement, a DHT22 sensor to measure temperature and humidity, and a BH1750 sensor to check light levels.

The collected data is sent to a web application built with Django and stored in a MySQL database. Administrators can view this information on a dashboard. This helps them monitor classroom occupancy, temperature, humidity, light levels, and device performance.

The system is built with the ESP32, IoT sensors, the Django framework, and MySQL databases. It can be deployed on cloud platforms like Render or AWS for remote monitoring and control. This solution aims to cut energy waste, improve classroom management, reduce operational costs, and create opportunities for more smart campus and energy management systems.

Technology Stack

Front-End: HTML, CSS, JavaScript

Back-End: Python, Django

Database: MySQL

IoT: ESP32, PIR, DHT22, BH1750, Relay

Cloud: Render / AWS

References

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[3] L. Monti et al., "Edge-Based Transfer Learning for Classroom Occupancy Detection in a Smart Campus Context," Sensors, vol. 22, no. 10, Art. no. 3692, 2022.

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